

Day 13: DAX Filter Context & CALCULATE() Mastery

📌 Definition to Remember

CALCULATE() is the single most powerful function in DAX. It is the only function that can modify, override, expand, or clear the existing **Filter Context** of a visual.

1. The Two Evaluation Contexts in DAX

1. **Row Context (Iterative)**: Knowing the values of the *current single row*. Present in calculated columns and iterator functions (**SUMX**).
2. **Filter Context (Analytical)**: The set of active filters applied by slicers, report pages, chart axes, and cross-visual selections.

2. Anatomy of CALCULATE()

```
CALCULATE(  
    <Expression / Measure>,  
    <Filter_Modifier_1>,  
    <Filter_Modifier_2>,  
    ...  
)
```

3. Modifying Filter Context Examples

```
// Example 1: Overriding Category Filter  
Audio Product Sales =  
CALCULATE(  
    [Total Sales],  
    Dim_Product[Category] = "Audio"  
)  
  
// Example 2: Multiple Filter Conditions (AND logic)  
US High Value Sales =  
CALCULATE(  
    [Total Sales],  
    Dim_Customer[Country] = "United States",  
    Fact_Sales[Quantity] > 5  
)
```

4. Context Transition

When a DAX measure is invoked inside a Row Context (like inside a calculated column or **SUMX**), DAX automatically initiates **Context Transition**, transforming the current row's key into an active Filter Context.

★ Must-Master Skills & Interview Points

1. **CALCULATE()** evaluates all filter arguments *in parallel* first, then evaluates the main expression under the newly modified filter context.
2. Explain Context Transition: how invoking a measure turns a row context into a filter context.
3. Why **CALCULATE** is the core building block of all advanced BI metrics.

⚡ Quick Recall

CALCULATE = Modifies Active Filter Context | Context Transition = Row Context → Filter Context